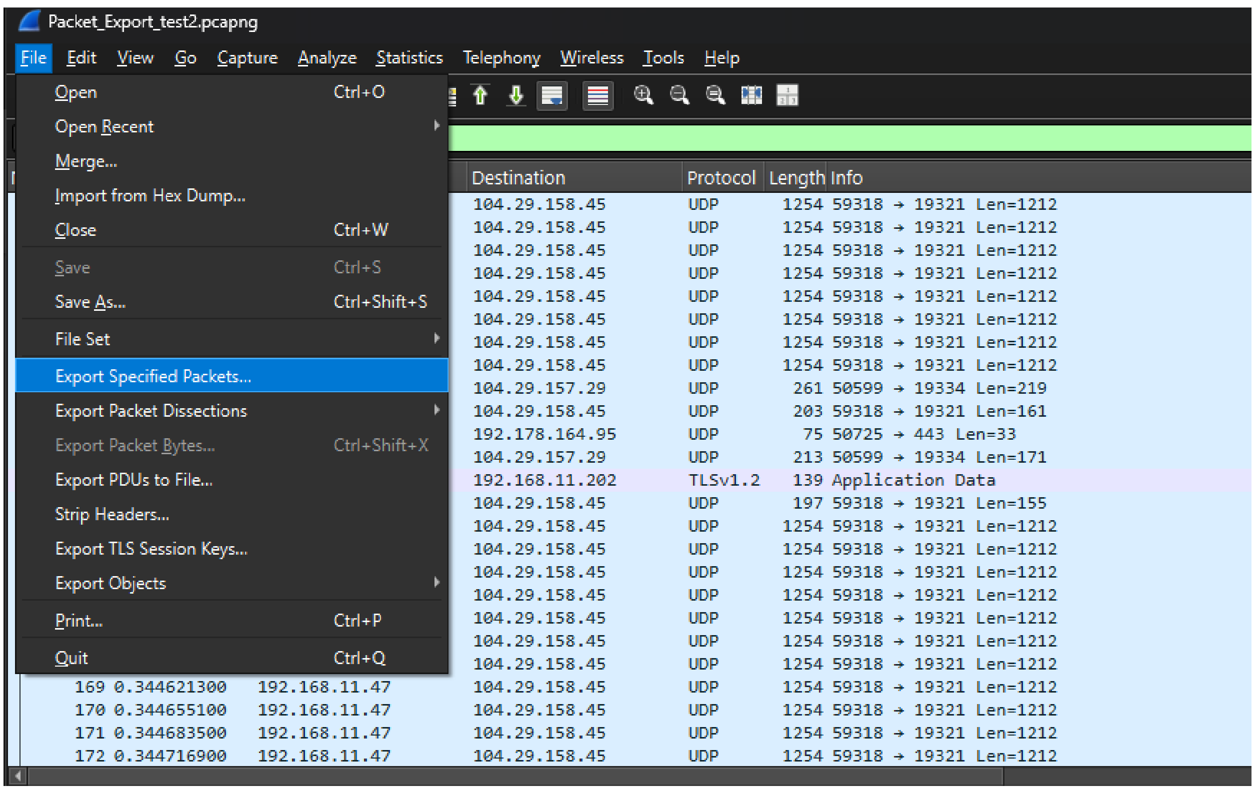
Wireshark Certification WCA-101

Notes from 25 Apr 2026

1.0 Utilize key features of Wireshark (10%)

- Open, save, and close files, and export specified packets.
  For exporting we needed to set a filter “ip.src==192.168.0.0/16” to capture the packets that we wanted to. Just selecting them, did nothing.

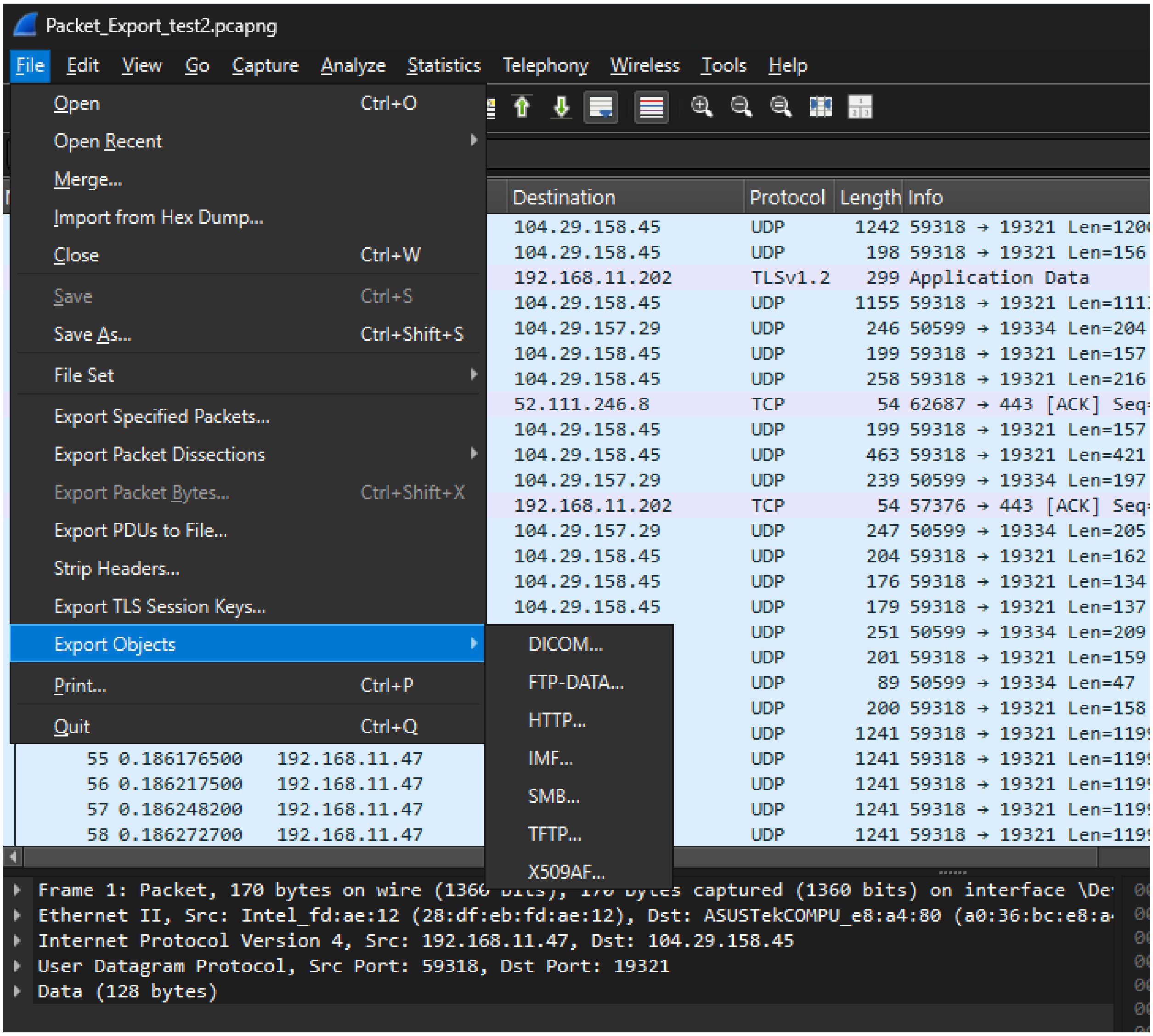
  This was a useful site: https://wiki.wireshark.org/DisplayFilters



- Describe the difference between different capture file formats, especially pcap and pcapng.
  Review: https://wiki.wireshark.org/Development/LibpcapFileFormat https://wiki.wireshark.org/libpcap https://medium.com/@pagaudiomonte/pcapng-vs-pcap-whats-the-diff-6670ca2273d3

  Major differences is that .pcap is more legacy (and what we pull from Linux boxes). .pcapng is the default now for captures in wireshark.

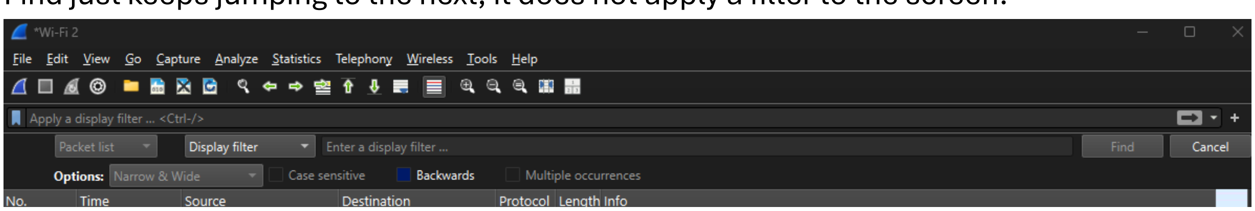
- Export objects from packet captures.



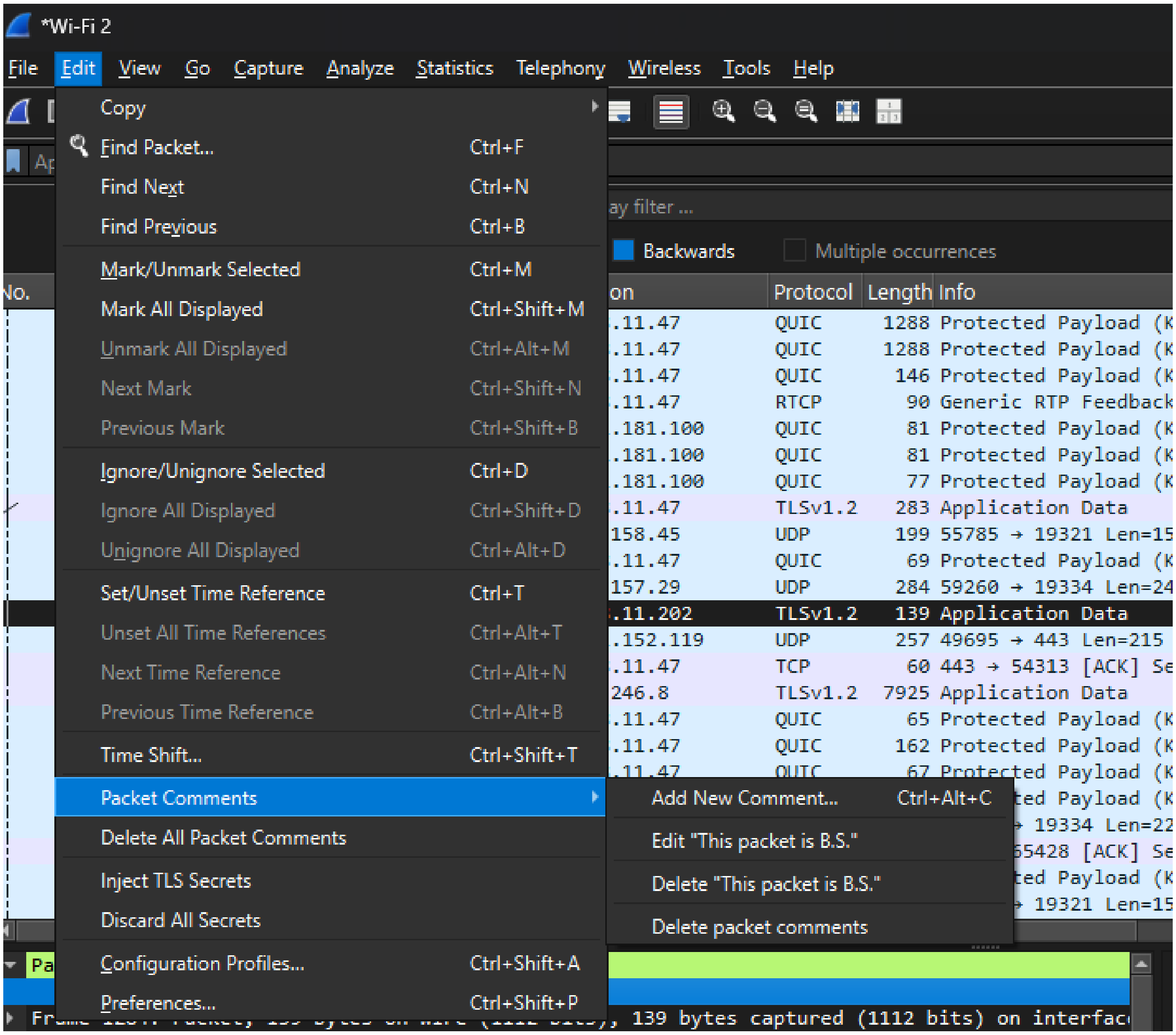
Read about the different file types: DICOM, FTP-DATA, HTTP, IMF, SMB, TFTP, X509AF

- Use 'Find Packet' to locate packets of interest.
  Edit -> Find Packet or “Ctrl +F”

  Find just keeps jumping to the next, it does not apply a filter to the screen.

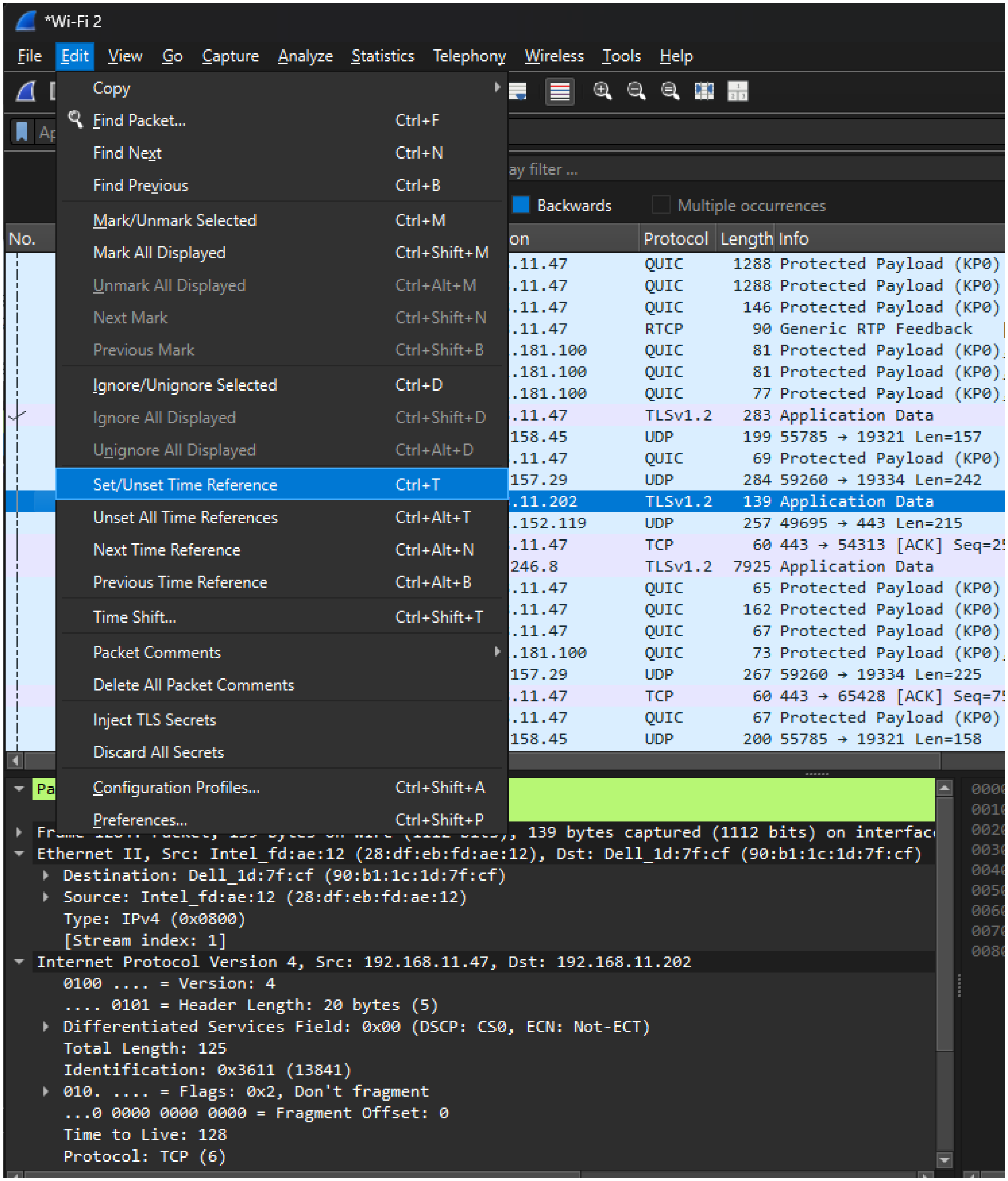


- Use the packet and file comments feature.



- Set/Unset time reference.
  From where you set this, time starts at “0”, like a zero day, or zero time from this packet etc.

  Once you have more than one (1), you can jump between them with Next and Previous time references.

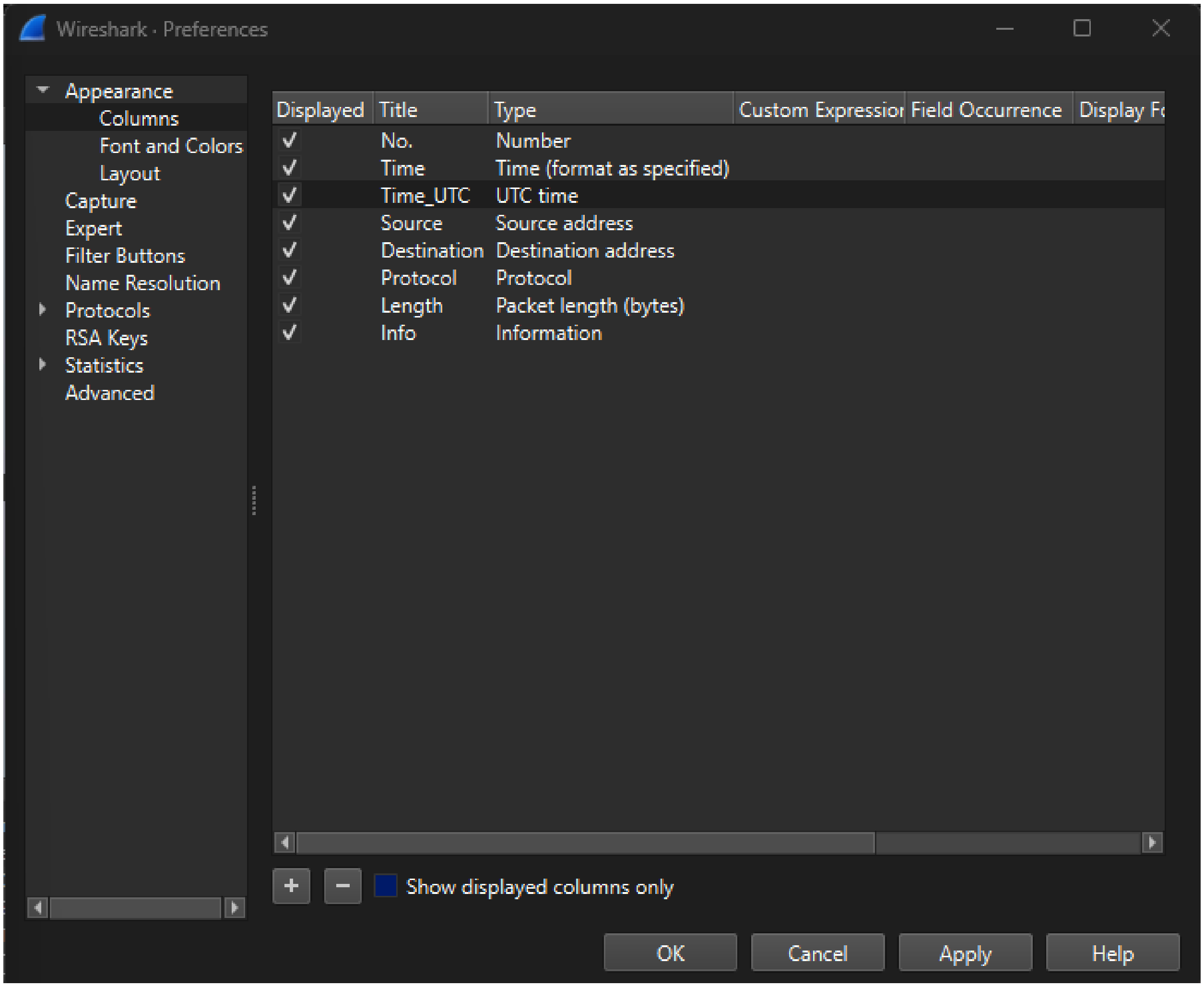


- Apply different time formats in a capture.

  Edit -> Preferences -> Columns

  Can apply any number of time formats

You can move it in the preferences tool, or out in the actual view of packets.



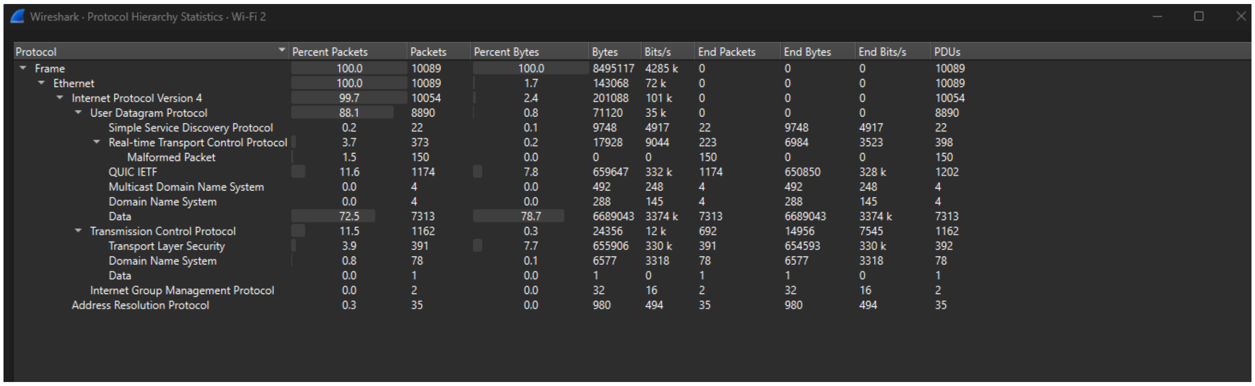
- Describe how Wireshark applies different name resolution options.
  It is unchecked by default, but we can set name resolution in edit -> preferences -> Name Resolution -> Resolve network (IP) addresses.
  - This is saved with the file in .pcapng files.

- Configure name resolution.
  Same as above, you can also set a column with name resolved instead of the default IP address.

- Use 'Decode as' feature.
  Applies to the packet that you are focused on.
  Analyze -> Decode As OR you can right click on a packet -> “Decode As”

- Use 'Capture File Properties' to identify key information about the capture.
  Statistics -> Capture File Properties

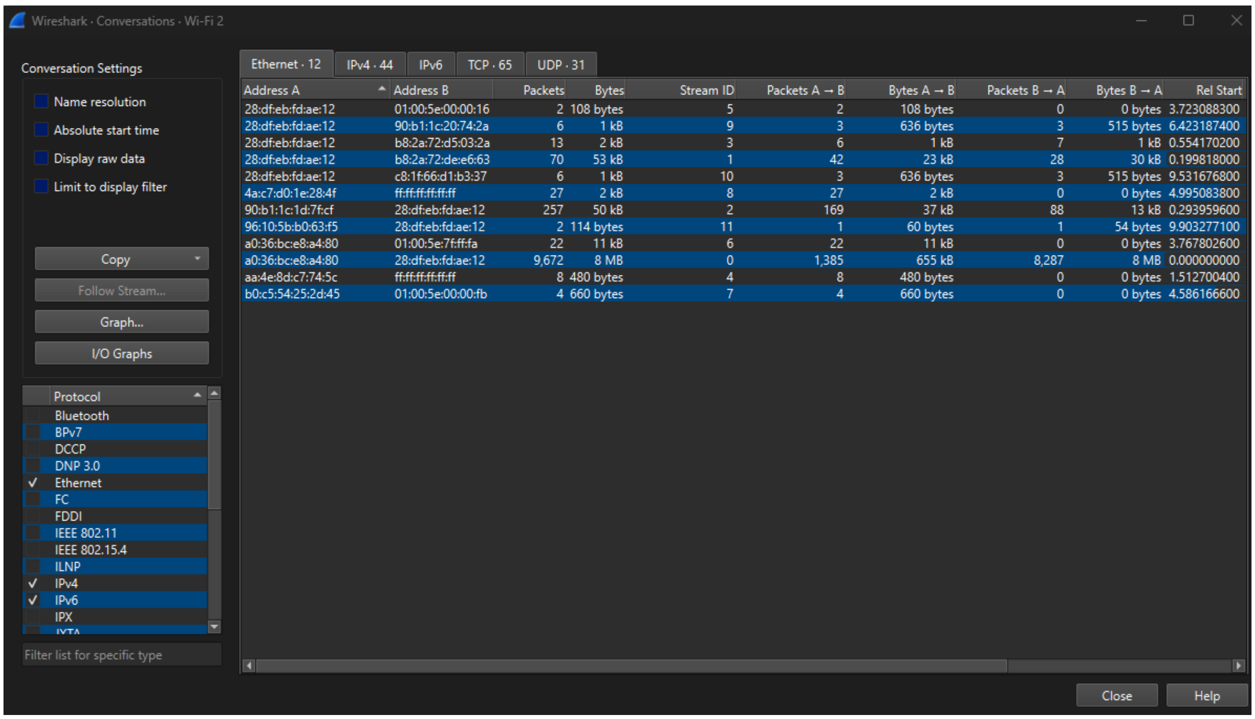
- Use 'Protocol Hierarchy' to identify key protocols in a capture.
  Statistics -> Procol Hierarchy
  Aggregates all the traffic up into percentages of overall traffic.



Wireshark · Protocol Hierarchy Statistics · Wi-Fi 2

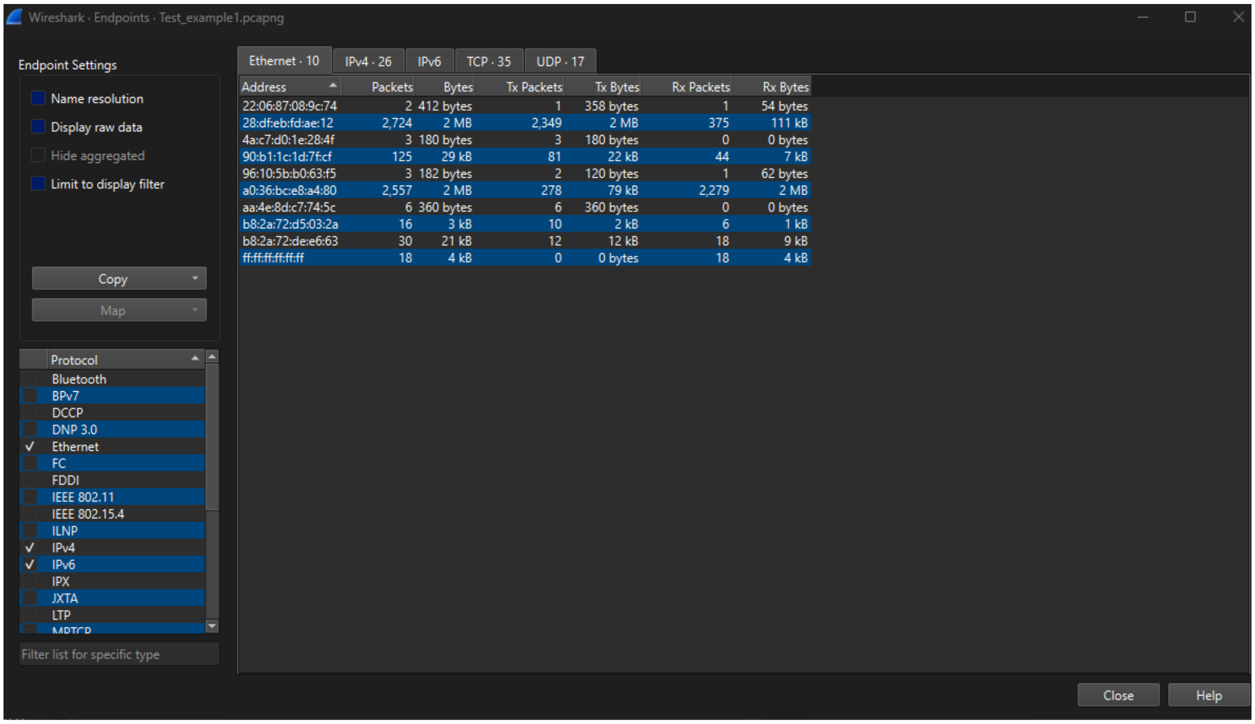
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	10089	100.0	8495117	4285 k	0	0	0	10089
Ethernet	100.0	10089	1.7	143068	72 k	0	0	0	10089
Internet Protocol Version 4	99.7	10054	2.4	201088	101 k	0	0	0	10054
User Datagram Protocol	88.1	8890	0.8	71120	35 k	0	0	0	8890
Simple Service Discovery Protocol	0.2	22	0.1	9748	4917	22	9748	4917	22
Real-time Transport Control Protocol	3.7	373	0.2	17928	9044	223	6984	3523	398
Malformed Packet	1.5	150	0.0	0	0	150	0	0	150
QUIC IETF	11.6	1174	7.8	659647	332 k	1174	650850	328 k	1202
Multicast Domain Name System	0.0	4	0.0	492	248	4	492	248	4
Domain Name System	0.0	4	0.0	288	145	4	288	145	4
Data	72.5	7313	78.7	6689043	3374 k	7313	6689043	3374 k	7313
Transmission Control Protocol	11.5	1162	0.3	24356	12 k	692	14956	7545	1162
Transport Layer Security	3.9	391	7.7	655906	330 k	391	654593	330 k	392
Domain Name System	0.8	78	0.1	6577	3318	78	6577	3318	78
Data	0.0	1	0.0	1	0	1	1	0	1
Internet Group Management Protocol	0.0	2	0.0	32	16	2	32	16	2
Address Resolution Protocol	0.3	35	0.0	980	494	35	980	494	35

- Use 'Conversations' to identify key communications in a capture.
  Statistics -> Conversations
  Shows overall frames between systems
  Shows data moving between systems



Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start
28:df:eb:fd:ae:12	01:00:5e:00:00:16	2	108 bytes	5	2	108 bytes	0	0 bytes	3.723088300
28:df:eb:fd:ae:12	90:b1:1c:20:74:2a	6	1 kB	9	3	636 bytes	3	515 bytes	6.423187400
28:df:eb:fd:ae:12	b8:2a:72:d5:03:2a	13	2 kB	3	6	1 kB	7	1 kB	0.554170200
28:df:eb:fd:ae:12	b8:2a:72:de:e6:63	70	53 kB	1	42	23 kB	28	30 kB	0.199818000
28:df:eb:fd:ae:12	c8:1f:66:d1:b3:37	6	1 kB	10	3	636 bytes	3	515 bytes	9.531676800
4a:c7:d0:1e:28:4f	ff:ff:ff:ff:ff:ff	27	2 kB	8	27	2 kB	0	0 bytes	4.995083800
90:b1:1c:1d:7f:cf	28:df:eb:fd:ae:12	257	50 kB	2	169	37 kB	88	13 kB	0.293959600
96:10:5b:b0:63:f5	28:df:eb:fd:ae:12	2	114 bytes	11	1	60 bytes	1	54 bytes	9.903277100
a0:36:bc:e8:a4:80	01:00:5e:7f:ff:fa	22	11 kB	6	22	11 kB	0	0 bytes	3.767802600
a0:36:bc:e8:a4:80	28:df:eb:fd:ae:12	9,672	8 MB	0	1,385	655 kB	8,287	8 MB	0.000000000
aa:4e:8d:c7:74:5c	ff:ff:ff:ff:ff:ff	8	480 bytes	4	8	480 bytes	0	0 bytes	1.512700400
b0:c5:54:25:2d:45	01:00:5e:00:00:fb	4	660 bytes	7	4	660 bytes	0	0 bytes	4.586166600

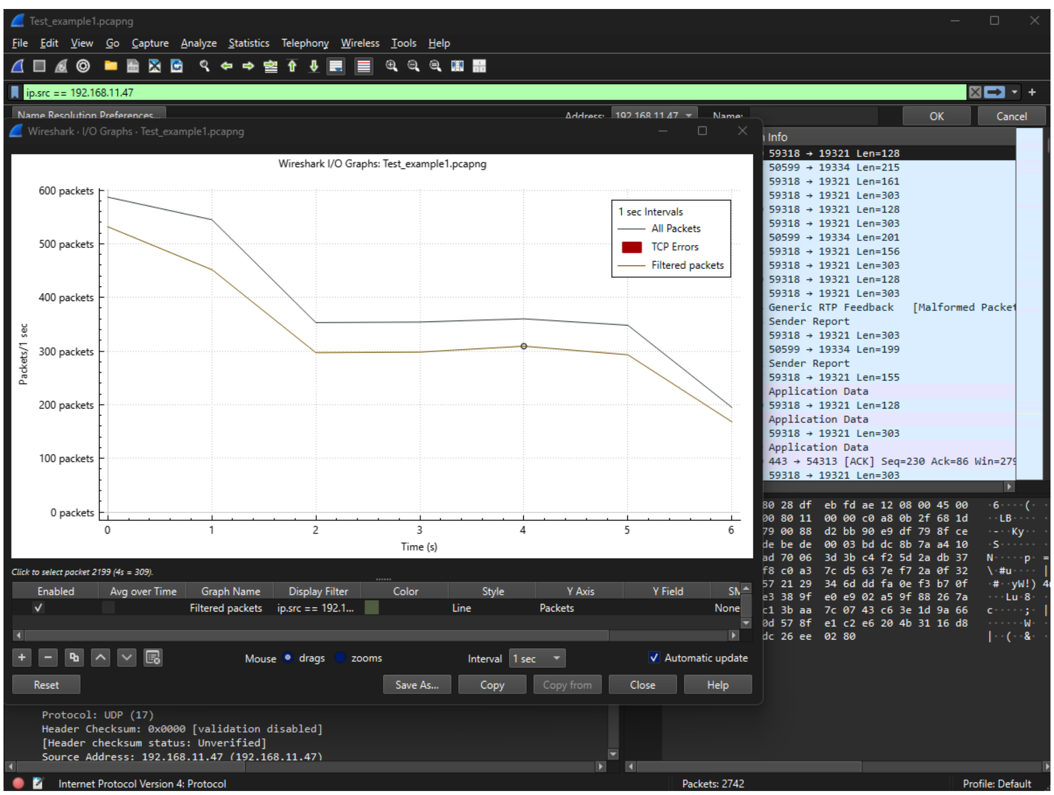
- Use 'Endpoints' to analyze host traffic in a capture.
  Statistics -> Endpoints



- Create and interpret an I/O graph that shows packets/s or bits/s for a given display filter.

  Set a filter "ip.src == 192.168.11.27"

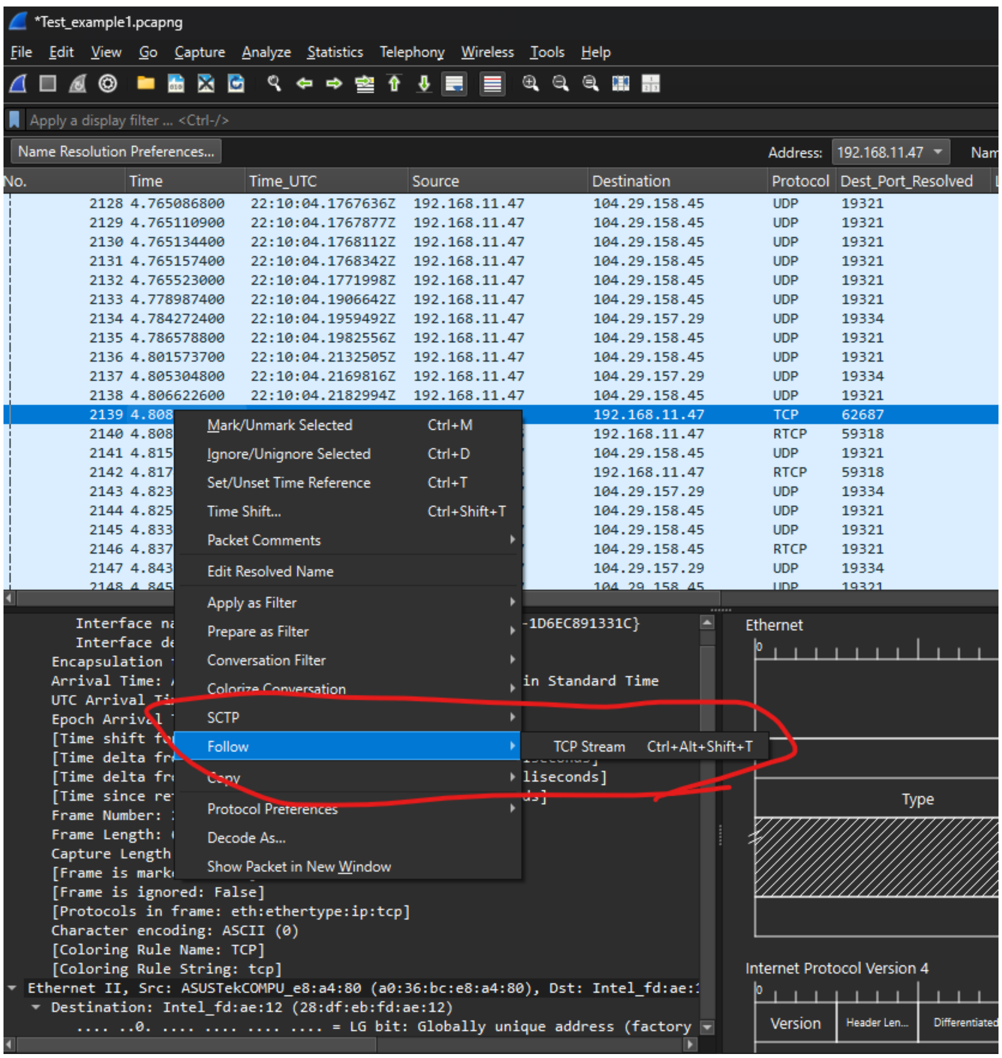
  Statistics -> I/O Graphs



- Distinguish between actual bytes captured and fields generated by the Wireshark
  Metadata is seen in [], in the frame information. It is not captured from the line.

- dissectors. (shown in [ ])
  Synthetic metadata in the packet details that is not actually captured at part of the packet. It's interpreted from the binary data.

- Use 'Follow TCP/UDP' stream.
  Right click on a TCP Stream -> Follow TCP/UDP stream and then click "Follow"



Future study for Section 1:

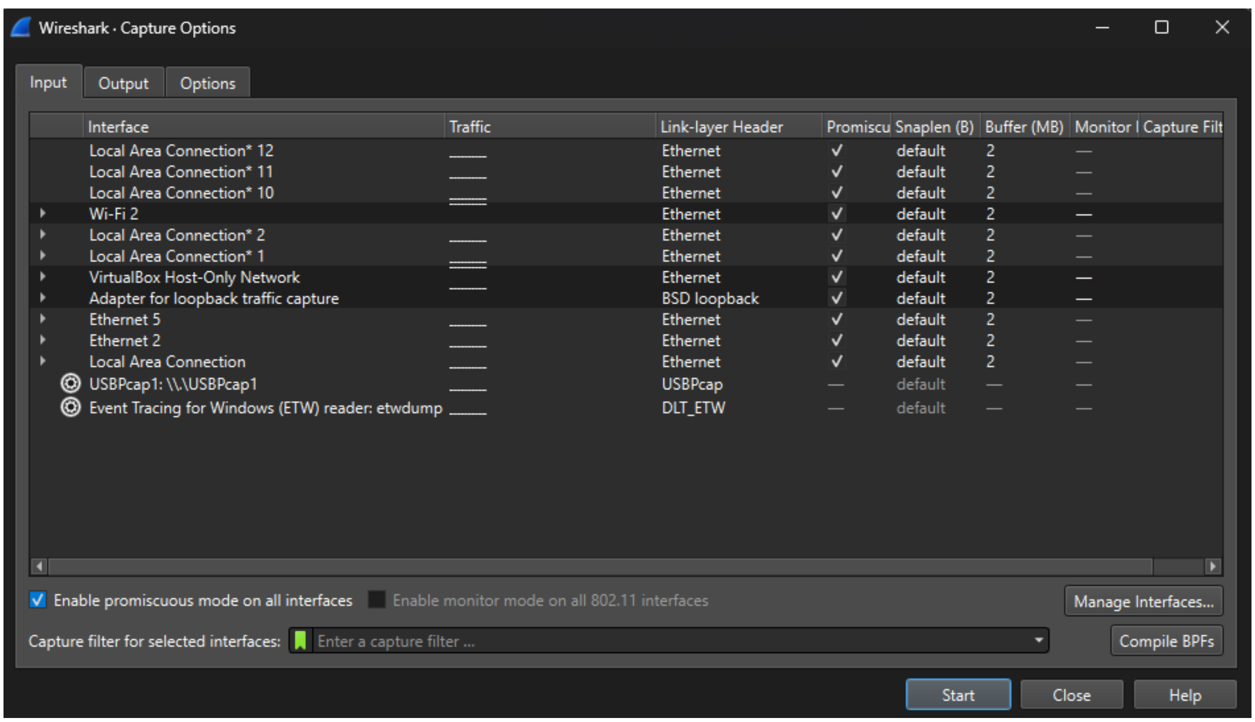
- https://gaia.cs.umass.edu/kurose_ross/online_lectures.htm

Notes from 2 May 2026

2.0 Utilize different methods of Capturing Traffic (10%)

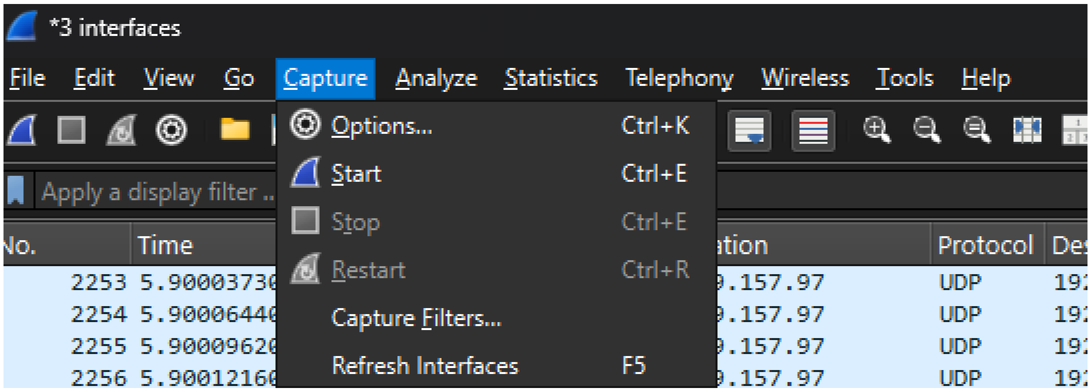
- Compare and contrast the benefits of different methods used for traffic capture. (Direct on Endpoint, Network Tap, Infrastructure Device, Port Mirror, Multi-Point Capture)
    - Direct on Endpoint is captured on the system you want to listen from.
    - Network Tap – Physical hardware that is between the switch and the device.
        - Can be undetected in the case of electromagnetic systems.
    - Infrastructure devices – Physical hardware between systems.
    - Multi-Point Capture - is the collection of many interfaces at the same time.

- Select the appropriate interface to capture traffic in Wireshark.
  Capture -> options
  Use “CTRL” to select multiple before capture.



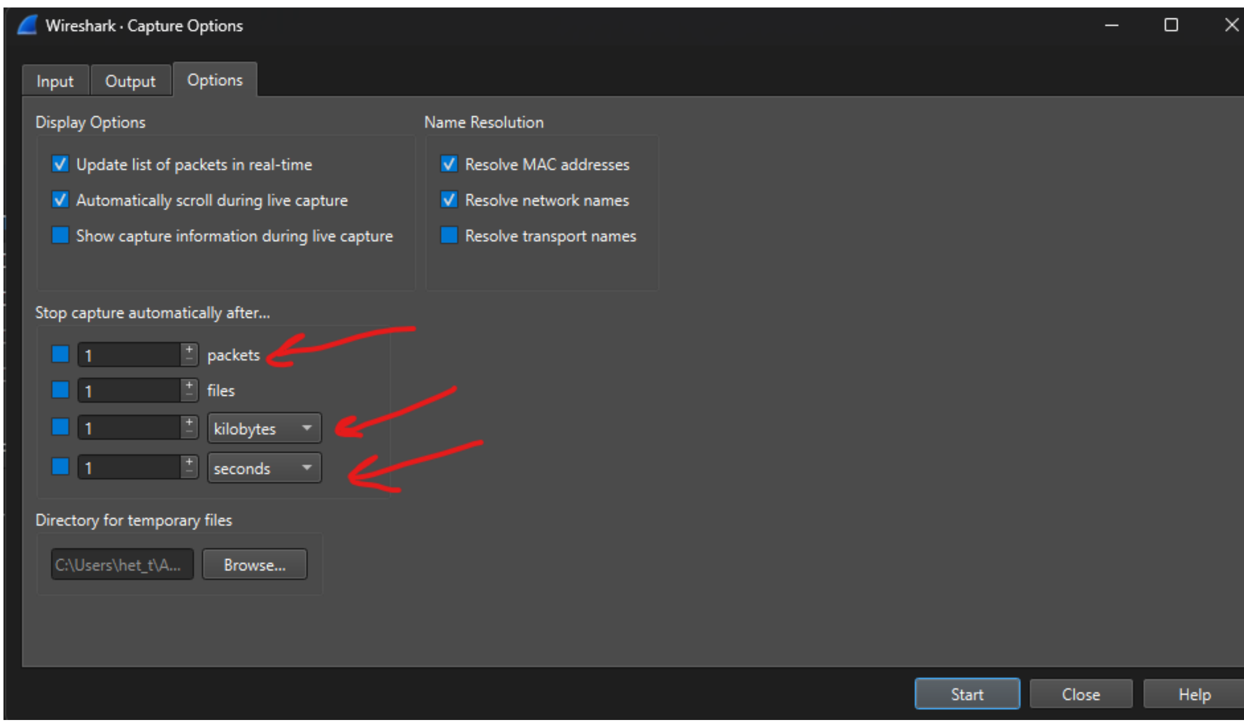
- Start/Stop/Restart Capture in Wireshark.

Capture -> Option you want to select

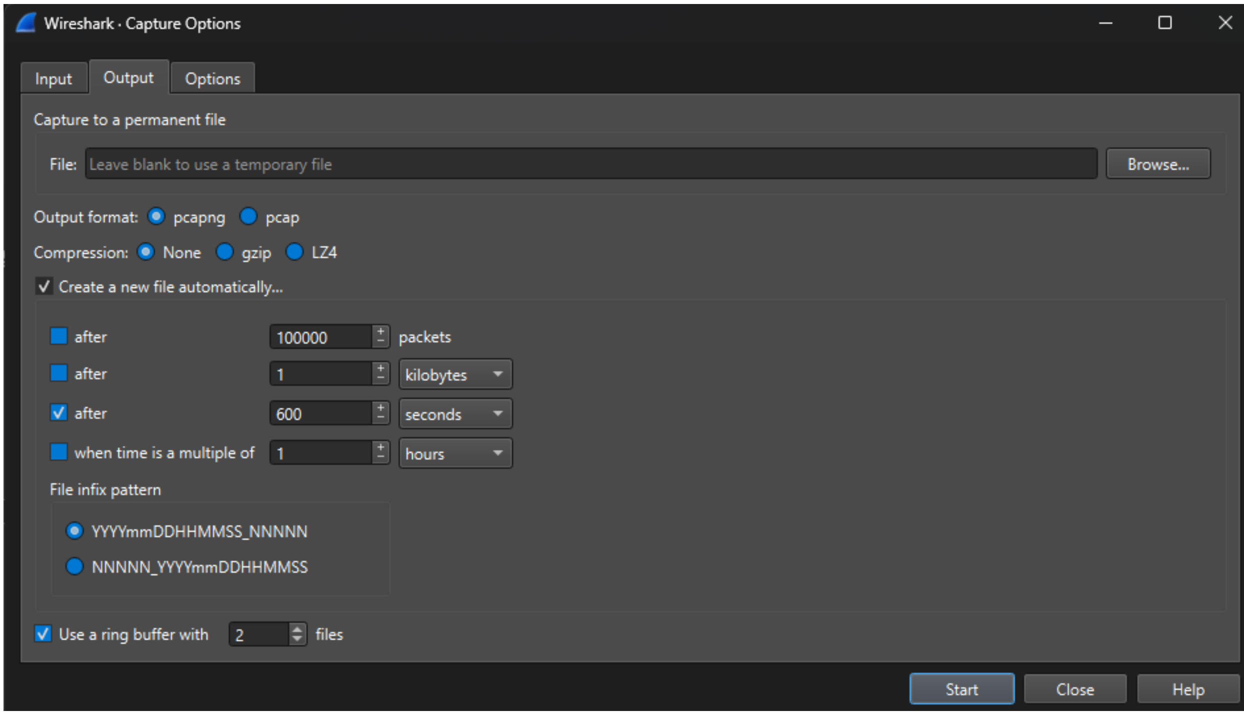


- Limit capture by file size, packets, or duration.
  - Wireshark

    Capture -> Options -> Tab Options



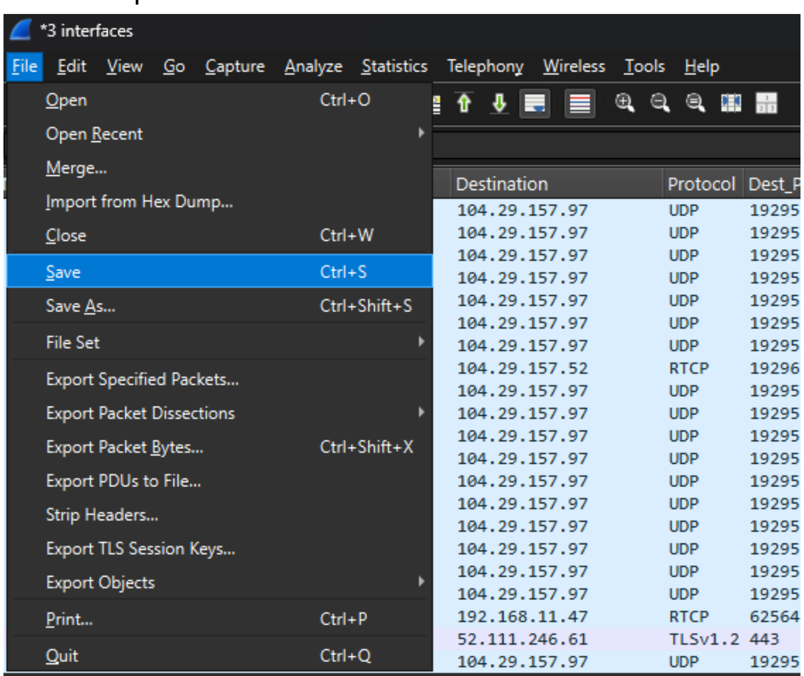
  - Tcpdump
    - By file size ‘-C’
    - By number of packets ‘-c’ or count
    - By duration of time ‘-G’
      - USE `tcpdump -h` to reference these.
- Implement a Ring Buffer.
  - Wireshark’s ring buffer is a capture mode for long-term monitoring that saves data across multiple, rotating files to prevent filling up disk space.
  - Steps – Capture -> Options -> Output Tab



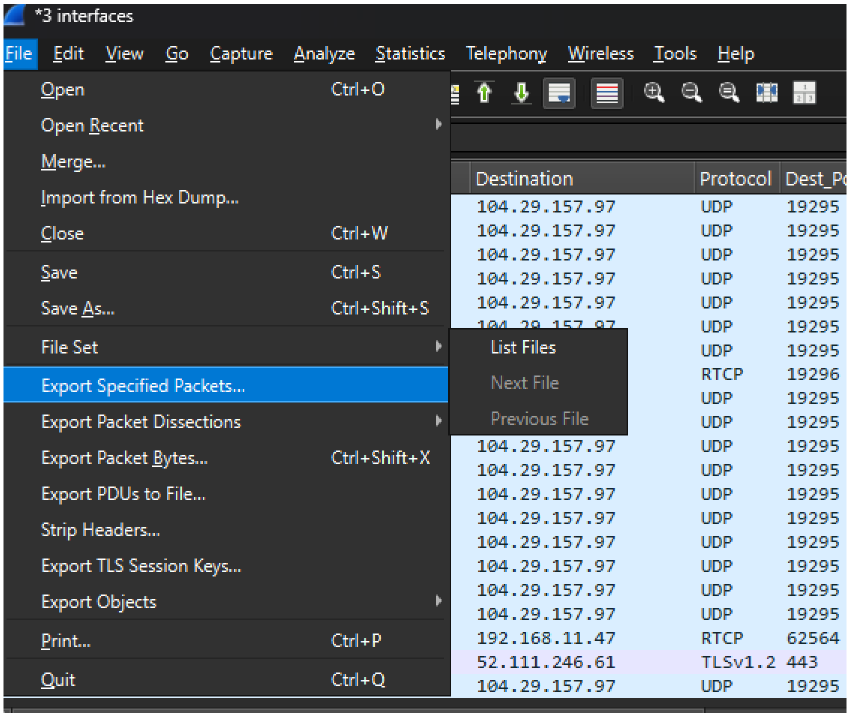
TCPDump Version:

- tcpdump -i <interface> -C <size_in_MB> -W 2 -w /path/to/cap_file

- Save a capture.



- Export specified packets to a new file.



Export only works on filtered files, otherwise you export everything.

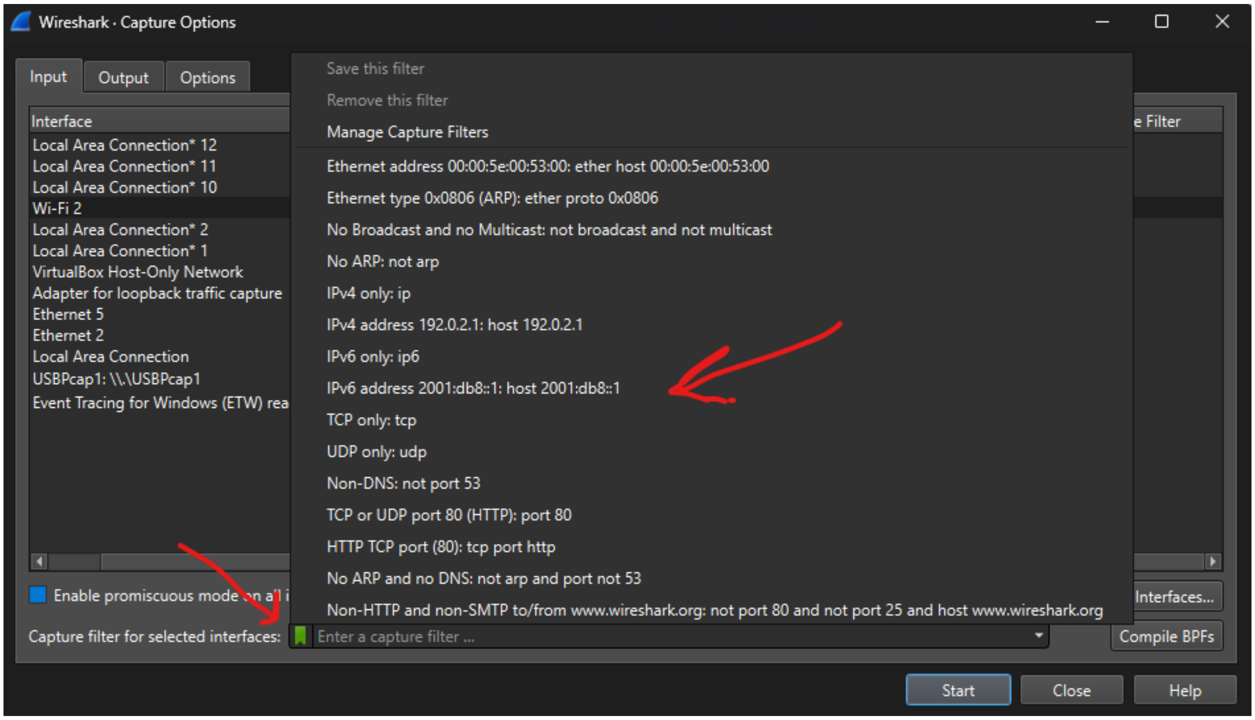
- Capture traffic using command line tools.
  Tcpdump in Linux - Probably not on your system for security reasons, ask admin.
  Tshark in Windows – Probably not in your base path, so move to correct directory.

- Describe the purpose of using promiscuous or monitor mode during a capture
  - **Promiscuous mode** captures all traffic on the *currently connected* network, even if addressed to other devices.
  - **Monitor mode** captures *all* wireless packets on a specific radio frequency, regardless of the network or access point, and is ideal for in-depth sniffing.

3.0 Filter traffic using capture and display filters. (12%)

- Compare and contrast Display Filters and Capture Filters.
  Capture filters (set *before* capturing) limit which packets are saved to disk, reducing file size and system load.
  Display filters (applied *after* capturing) hide uninteresting traffic in the current view without deleting it, allowing for detailed, deep-packet inspection.
- Implement a Capture Filter to capture only traffic from a single protocol, IP address,

MAC address, or port (range).



- Use multiple methods to create a Display Filter to isolate traffic for a single protocol or a property of a protocol. (manual entry, right click, drag/drop)

1. Type it in at the top.
2. You can right click on the actual row and column, to select filter.
3. You can select the information in the bottom left corner and right click to apply the filter.
    a. This can also be dragged right into the top display filter field and it will apply that filter to all display traffic.

- Use membership filters (tcp.port in {80,443})

https://www.wireshark.org/docs/wsug_html_chunked/ChWorkBuildDisplayFilterSection.html

- Use logical operators to connect multiple filters together.

Table 6.7. Display Filter Logical Operations

English	C-like	Description	Example
and	&&	Logical AND	`ip.src==10.0.0.5 and tcp.flags.fin`
or	||	Logical OR	`ip.src==10.0.0.5 or ip.src==192.1.1.1`
xor	^^	Logical XOR	`tr.dst[0:3] == 0.6.29 xor tr.src[0:3] == 0.6.29`
not	!	Logical NOT	`not llc`
[...]		Subsequence	See "Slice Operator" below.
in		Set Membership	`http.request.method in {"HEAD", "GET"}`. See "Membership Operator" below.

- Create a button for easy access to a Display Filter.

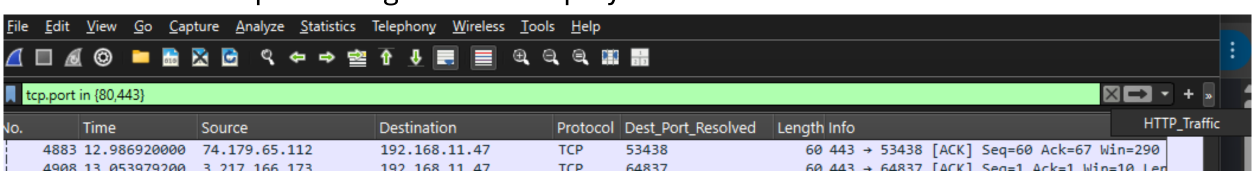
    Type in your display filter

    Click the button on the far right

      Give it a name and a description

      Hit "OK"

    Button will show up to the right of the Display



- Identify situations where a Display Filter will show incomplete or excess results (i.e. filter for HTTP, but do not see the TCP handshake)

    Incomplete

      HTTP (as above)

      RTP (will not show Session Initiation Protocol above it)

      Too many && (and)s

    Excessive

      Too many || (or)s will expand too much

- Identify the behavior of using ! (not) in different parts of filter logic by explaining the implicit 'any' and 'all' qualifiers.

The meaning of != (all not equal) was changed in Wireshark 3.6. Before it used to mean "any not equal".

All Not (!=, all_ne): Matches if none of the fields in a packet meet the condition.Example: tcp.port != 80 (Shows packets where neither source nor destination port is 80).

Any Not (!~, any_ne): Matches if at least one field in a packet does not meet the condition. (Note: The all_ne is typically used for the logical "all not").

- Apply filters from Statistics > Conversations and Statistics > Endpoints.

  Apply the filters in those locations and it will be in the display filter out in the normal panel.

- Create a filter using generated fields in Wireshark.

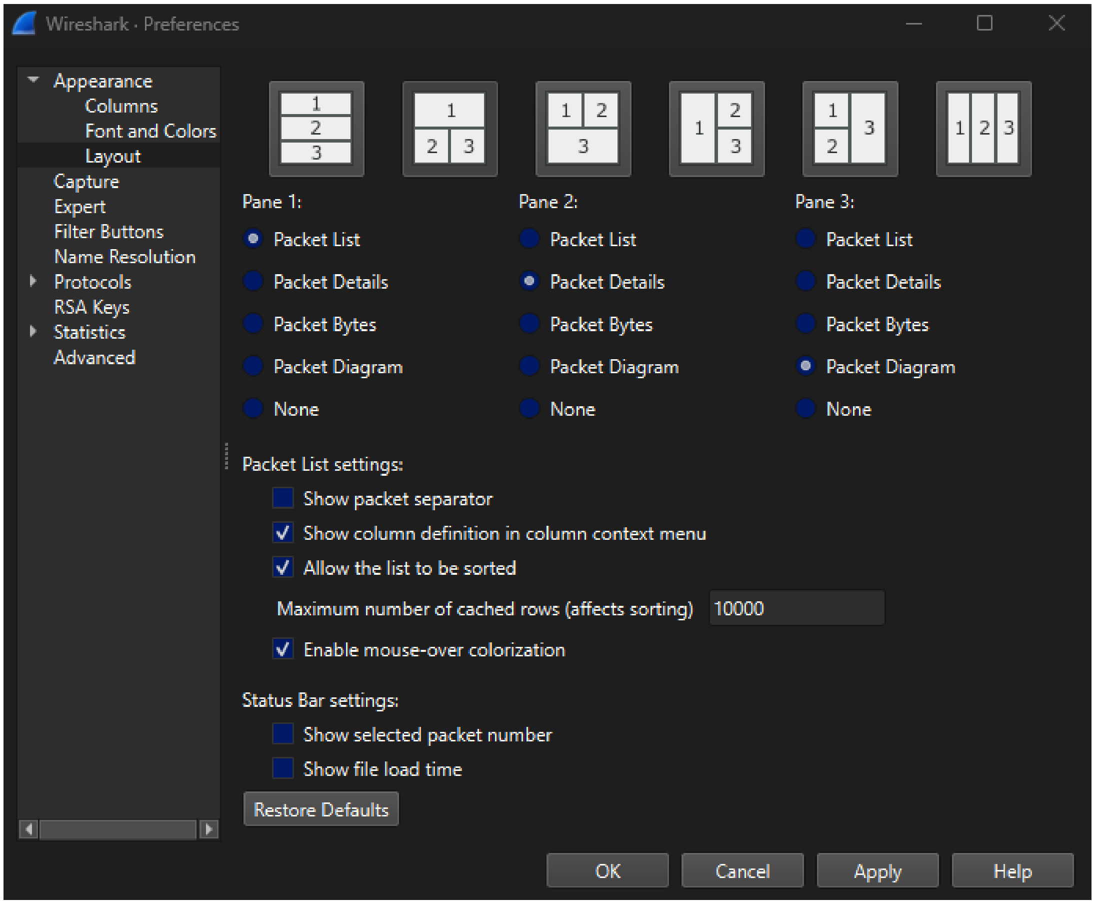
  Drag it up from the bottom left panel and put it in the display filter.

Notes from 9 May 2026

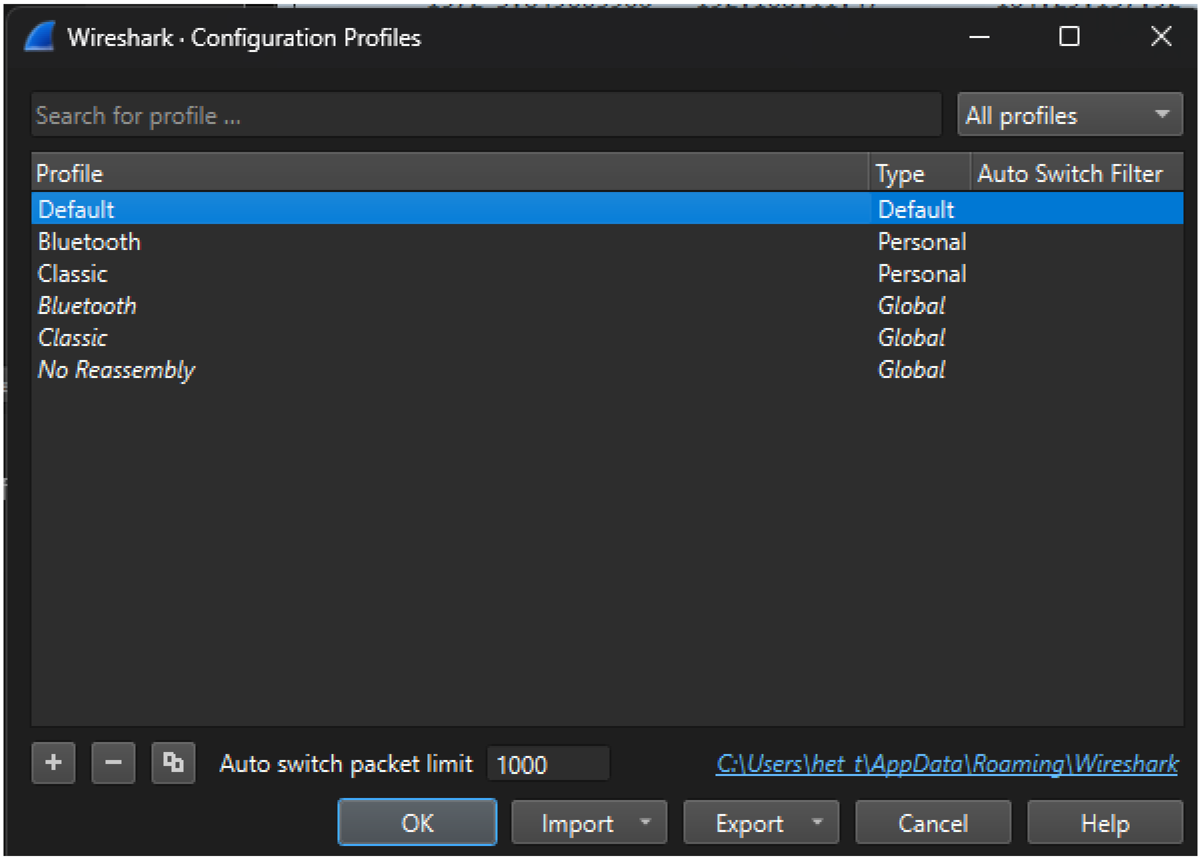
4.0 Configure, adapt, and use the Wireshark interface for different

scenarios. (5%)

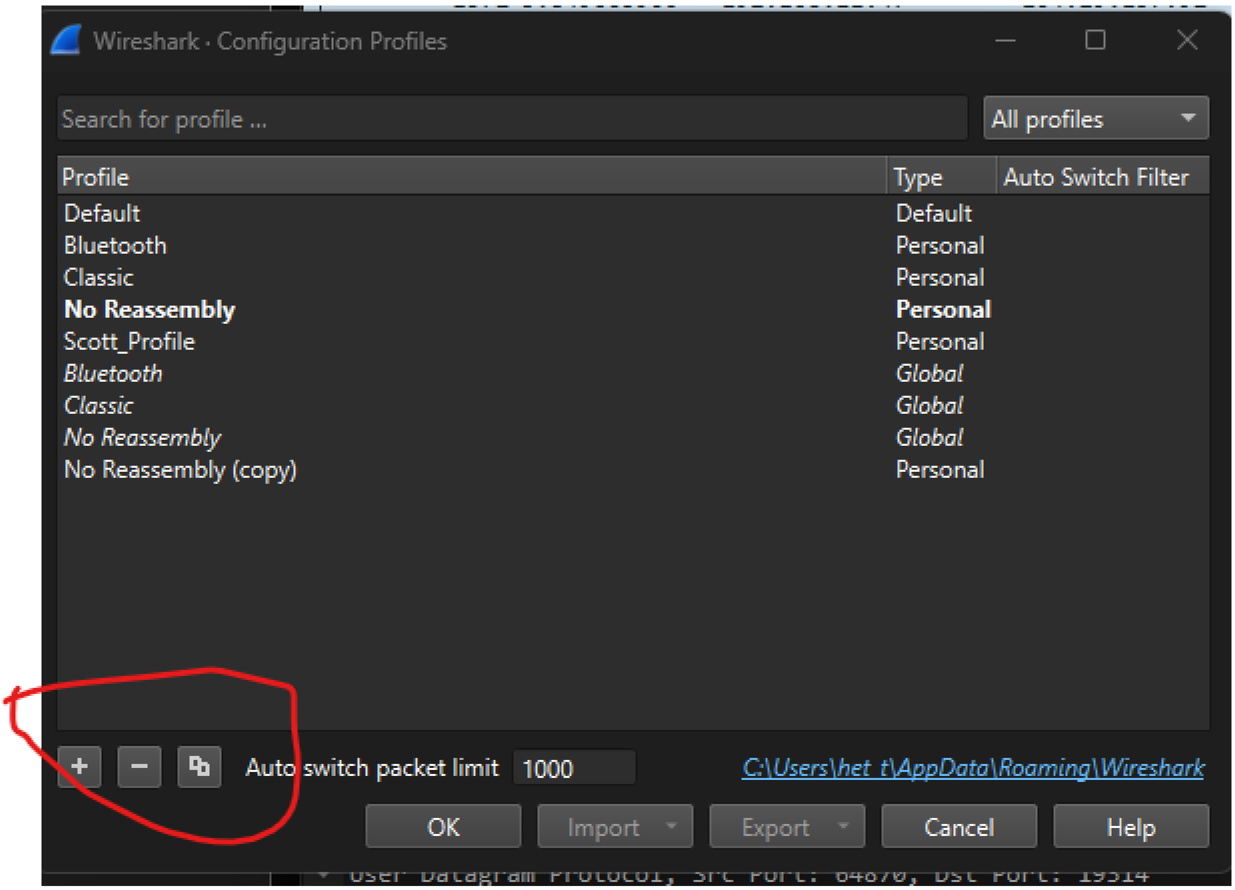
- Identify key components of the GUI (packet list, hex view, packet details, etc.).
    - Pick this in the Edit -> options -> Layout
- Modify panes with a different layout/features.
    - Same as above



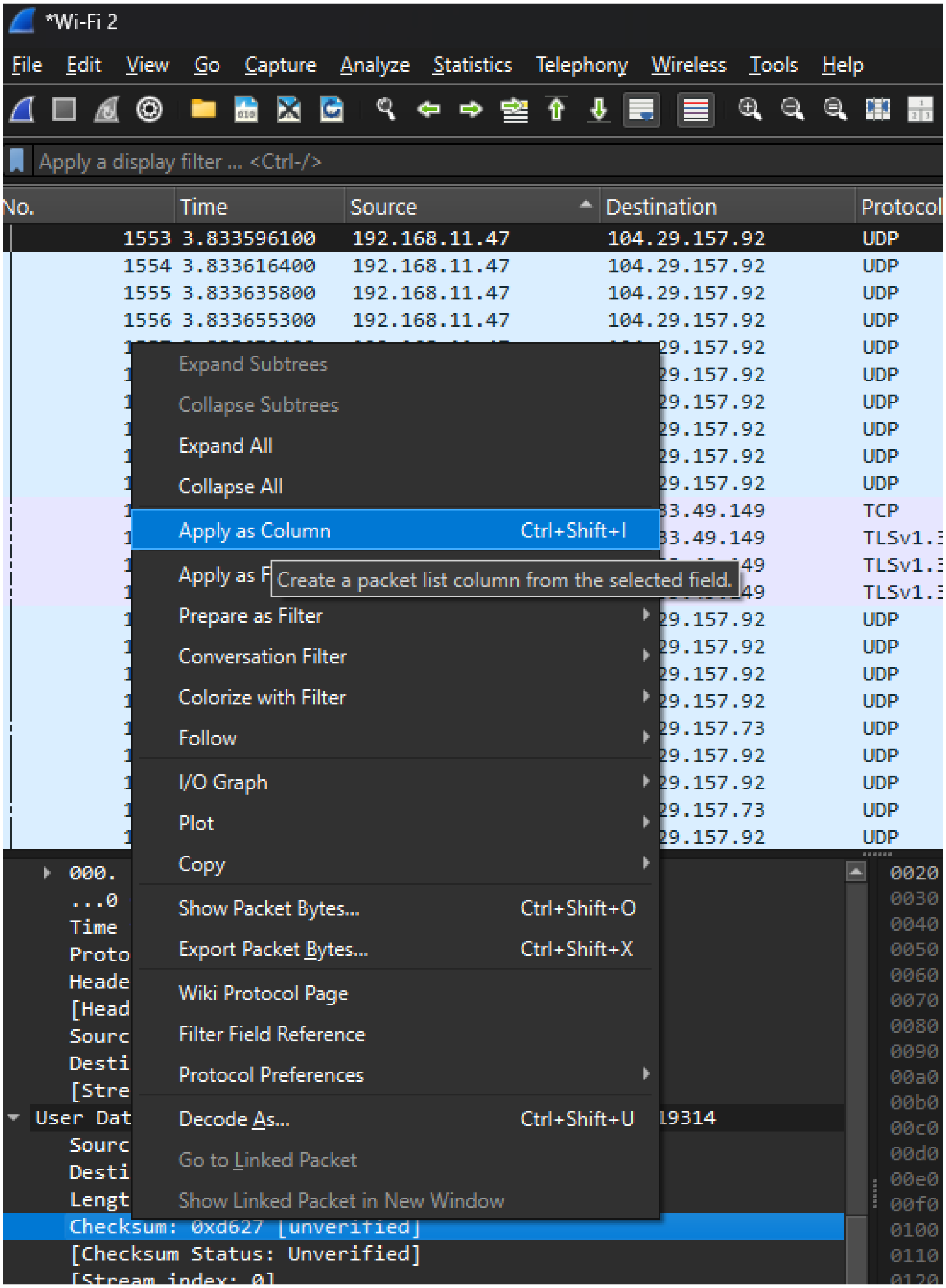
- Describe the value of using profiles.
    - Saves time giving you the views and window panes that you like to look at for certain traffic.
        - Profiles can be
            - Default
            - Personal
            - Global



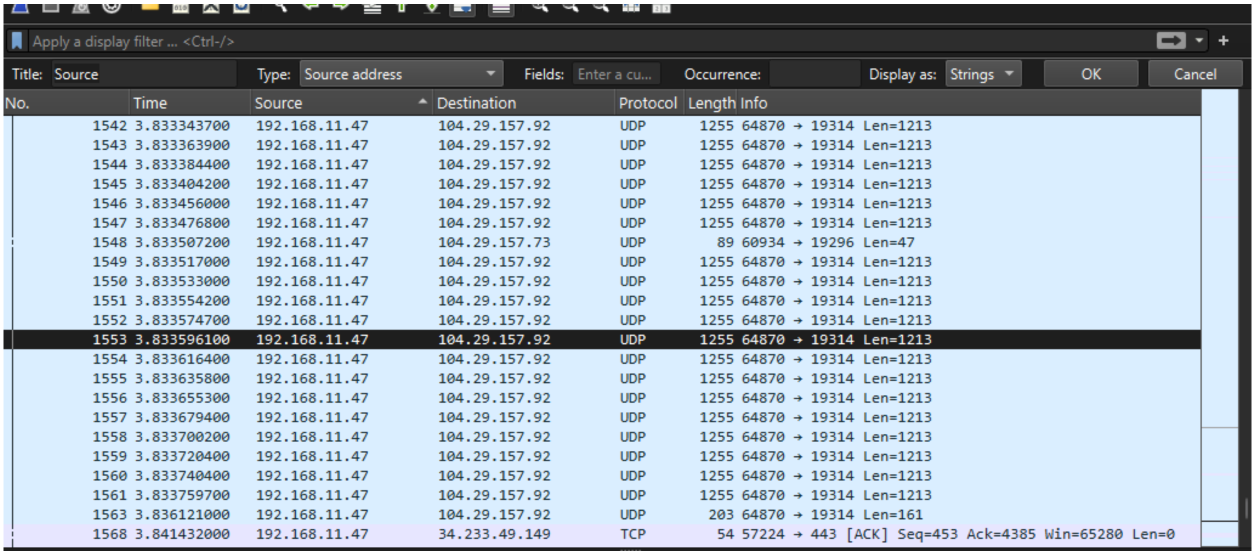
- Create/modify/copy a profile.



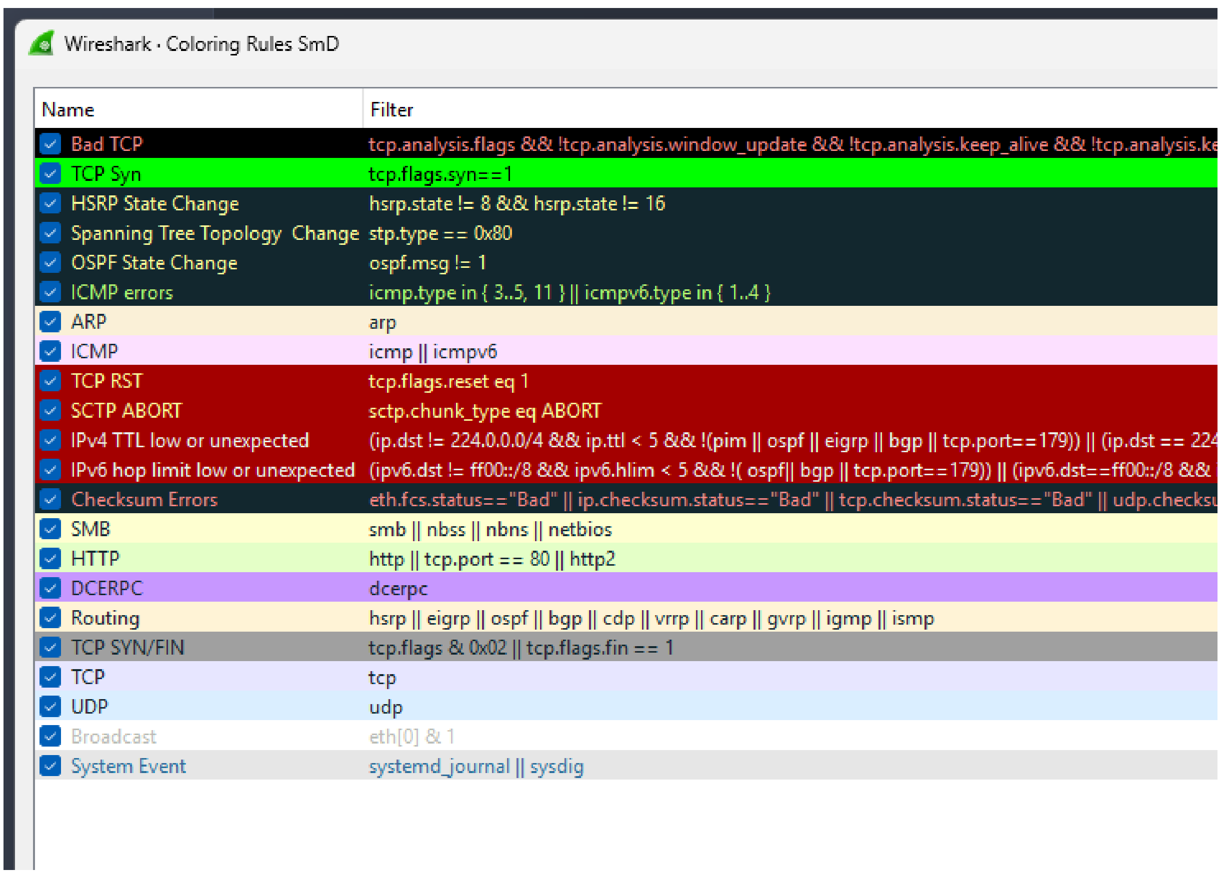
- Describe the importance of columns in troubleshooting.
  1. I know the data.
  2. I can filter or sort the data.
- Use multiple methods to add a column.
  - Select from the packet view and then “apply as column”



Right click on a column header and click “Edit Column”



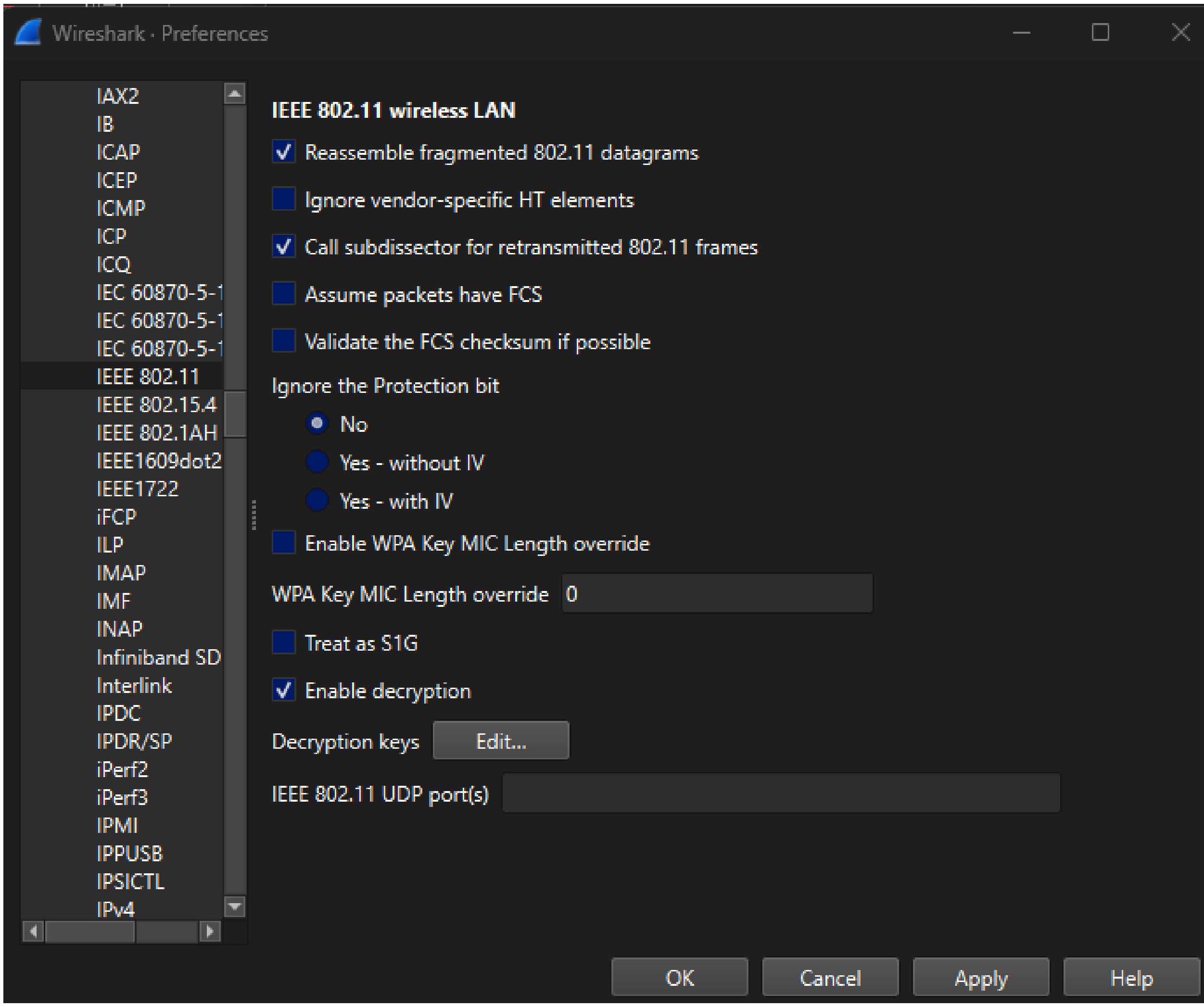
- Use coloring rules to highlight packets.
  - There are defaults (below)
  - There are also custom rules you can apply to conversations.



- Use the minimap (colored sidebar) to quickly locate packets of interest.
  - Right side shows a complete view of the entire capture, to jump around.
- Use the 'Colorize Conversation' feature.

- Right click on a packet
- Or view menu to colorize a conversation.

- Understand the importance of protocol preferences in your analysis.
  - Edit -> Preferences -> Protocols
    - Then you can set defaults for different protocols based on how they are set in your environment.



- Use the mark/unmark packet feature.
  Time shifts to 0.00000 related to each mark of time shift.

Notes from 16 May 2026

5.0 Identify and explain common network protocols dissected by

Wireshark. (43%)

(note: TCP is 17% of exam and is scored separately. Non-TCP protocols are 27% of exam)

5.1 ETHERNET

• Identify Fields of Ethernet frame.

• Describe minimum and maximum frame sizes.

• Explain why the CRC is missing from Ethernet frame in Wireshark.

• Identify common Ethertypes (IPv4, IPv6, ARP).

• Distinguish between Unicast, Broadcast, and Multicast MAC addresses.

• Describe how the frame header is modified to accommodate VLAN tags.

5.2 ARP

• Describe the purpose of the ARP protocol.

• Identify and explain the purpose of different types of ARP packets.

• Create filters for different types of ARP traffic.

• Describe the difference between a broadcast ARP and a unicast ARP.

5.3 IPv4

• Describe common features of the IP protocol.

• Describe header values of the IP protocol (TTL, Fragmentation, Packet Length, Protocol ID, IP ID).

• Describe public, private, multicast and APIPA IP address ranges.

• Describe how NAT works and why this should be considered when capturing and analyzing network traffic.

• Identify and explain the purpose of the IP TTL field.

• Predict most likely distance, in hops, from the capturing device.

• Describe different IP identification strategies and how to use them in troubleshooting.

5.4 ICMPv4

• Identify ICMP message types and their purpose.

• Identify the IP packet which triggered an ICMP error message or reply.

5.5 IPv6

• Identify types of IPv6 addresses (Link Local, Global Unicast, Multicast).

5.6 ICMPv6

• Identify and explain components of neighbor/router discovery protocol.

• Identify and explain purpose of Neighbor Advertisements/Solicitations.

Notes from 23 May 2026

5.0 Identify and explain common network protocols dissected by

Wireshark. (continued)

5.7 UDP

• Identify UDP traffic.

• Identify higher layer protocols which use UDP.

• Describe the UDP stream id and conversation timestamps.

• Describe why UDP is used in multicast or broadcast IP traffic.

5.8 DHCPv4

• Identify and describe the 4 phases (DORA).

• Describe the different purposes of a 'DHCP Request' message.

• Identify DHCP options and parameters (router, dns, subnetmask, custom options).

• Describe APIPA and how it relates to DHCP.

5.9 DNS

• Identify DNS requests and replies.

• Use DNS information to identify relevant traffic.

• Distinguish between different DNS record types.

5.10 TCP (17%)

• Describe the components of a 3 way handshake.

• Describe different methods TCP session are torn down.

• Explain how iRTT is measured.

• Describe and calculate Maximum Segment Size(MSS).

• Describe how TCP flags are used to establish and tear down a TCP session.

• Explain how sequence and acknowledgment numbers are used to maintain reliability.

• Explain TCP options and their purpose.(EOL, NOP, MSS, SACK, DSACK, Window Scaling)

• Describe the purpose of a DUP ACK.

• Identify the byte range of missing segments based on a DUP ACK with SACK or DSACK.

• Describe what each line represents in a TCP Stream Graph.

• Describe the purpose of TCP reassembly in Wireshark.

• Describe the TCP stream id and conversation timestamps.

Notes from 30 May 2026

6.0 Use Wireshark to troubleshoot common issues with protocols

listed above. (20%)

• Determine network topology only using information in a packet capture.

• Perform TCP sequence and acknowledgment number analysis.

• Distinguish server performance from slow transfer times (for instance in HTTP).

• Identify the effect of high RTT on request/response protocols. (For example: HTTP, SMB, SQL)

• Identify the effect of a low window size in combination with high RTT.

• Given a capture, identify potential network communication issues using information from ARP, DHCP, and ICMP.

Additional Study Materials